

Bill of Materials

AS20P2_1_3

Project: AS20P2_1_3.PrjPcb
Variant: None

Report Date: 23/09/2025



#	Value	EFootprint	Designator	Description	Note	MAG_CODE	Layer	Mount	Quantity
1	100n	E0603C	C1, C11, C13, C18, C19, C21, C22, C37, C52, C76, C77, C79, C94, C101, C106, C109, C136, C157, C159, C160, C166, C168, C179, C182, C183, C184, C188, C199, C200, C236, C240, C246	Capacitor 0603, 50V		CO-CSMD100N50VX7R	Top	SMD	32
2	100n	E0603C	C2, C6, C8, C12, C14, C16, C23, C24, C26, C29, C32, C34, C35, C36, C40, C45, C54, C55, C60, C61, C62, C63, C64, C65, C66, C67, C68, C69, C70, C71, C74, C75, C78, C81, C82, C83, C84, C85, C86, C87, C88, C89, C90, C91, C92, C93, C95, C97, C98, C99, C100, C102, C103, C104, C107, C108, C110, C111, C112, C113, C114, C115, C116, C117, C118, C119, C120, C121, C137, C138, C143, C145, C146, C147, C150, C152, C155, C156, C161, C163, C164, C165, C167, C169, C170, C171, C172, C173, C174, C175, C176, C180, C181, C185, C186, C187, C189, C216, C217, C218, C219, C237, C241, C244, C249	Capacitor 0603, 50V		CO-CSMD100N50VX7R	Bottom	SMD	105
3	NM	E0603C	C3, C5, C51, C59, C220, C221, C222, C223, C225, C226, C227, C228, C229, C230, C231, C233, C234, C235	Capacitor 0603, 50V, Capacitor 0603, Capacitor 0805, 50V		NM	Top	SMD	18
4	100n	E0603C	C4, C72, C129	Capacitor 0603, 25V		CO-CSMD100N25VX7R	Top	SMD	3
5	10u	E0603C	C7, C31, C41, C42, C44, C49, C56, C80, C96, C122, C123, C127, C139, C140, C148, C248, C251	Capacitor 0603, 10V		CO-CSMD10MF0603	Bottom	SMD	17
6	10u	E0603C	C9, C10, C15, C20, C25, C30, C33, C38, C39, C46, C48, C50, C53, C57, C58, C105, C124, C141, C142, C149, C245, C247, C250	Capacitor 0603, 10V		CO-CSMD10MF0603	Top	SMD	23
7	100n	E0603C	C17, C73, C130, C131, C132, C133, C134, C135, C153, C154, C201, C202, C203, C204, C205, C206, C207	Capacitor 0603, 25V		CO-CSMD100N25VX7R	Bottom	SMD	17
8	47u	E1210C	C27, C211	Capacitor 1210, 16V		CO-CSMD47MF121016V	Top	SMD	2
9	470n	E0603C	C28, C47	Capacitor 0603, 16V		CO-CSMD470N0603X7R16	Bottom	SMD	2
10	10u	E1206C	C43, C212	Capacitor 1206, 25V		CO-CSMD10MF120625X5R	Top	SMD	2
11	470n	E0603C	C125, C126, C252	Capacitor 0603, 16V		CO-CSMD470N0603X7R16	Top	SMD	3
12	1u	E2220C	C128, C198	Capacitor 2220, 250V		CO-CSMD22201MF250	Top	SMD	2
13	1u	E2220C	C144, C208	Capacitor 2220, 250V		CO-CSMD22201MF250	Bottom	SMD	2
14	10u	E1206C	C151	Capacitor 1206, 25V		CO-CSMD10MF120625X5R	Bottom	SMD	1
15	47u	E0805C	C158, C162	Capacitor 0805, 10V		CO-CSMD47MF0805	Bottom	SMD	2
16	NM	E1206C	C224, C232	Capacitor 0603, 50V		NM	Top	SMD	2
17	100p	E0603C	C253	Capacitor 0603, 50V		CO-CSMD100P060350V	Top	SMD	1
18	B260	SMB_UNI	D1, D2, D7, D8, D21, D22	Schottky Barrier Diode, B260-13-F		DI-B260-13-F	Top	SMD	6
19	MBR-0530	SOD123AK	D3, D4, D5, D6	Schottky Diode MBR0530		IC-MBR0530	Top	SMD	4
20	MBR-0530	SOD123AK	D9, D10, D12, D14	Schottky Diode MBR0530		IC-MBR0530	Bottom	SMD	4
21	1A25	FUSE_MF-SM100	F1, F2	Littelfuse PTC Resettable Fuse		FU-SMD125F-2	Top	SMD	2
22	CON5	STRIP5-F	FIL_A-IN, FIL_B-IN	Connector, 5 Pin, Female, Pitch 2.54mm		CN-80187050-1001	Top	PTH	2
23	CON3		FIL_A-OUT, FIL_B-OUT	Strip Connector, 3 Pin, Female, Pitch 2.54mm		CN-80187050-1001	Top	PTH	2
24	CON6	STRIP6	J1	Strip Connector, 6 Pin, Male, Pitch 2.54mm		CN-STRL5169140231116	Top	PTH	1
25	CON4	STRIP4	J2, J8	Strip Connector, 4 Pin, Male, Pitch 2.54mm		CN-STRL5169140231116	Top	PTH	2
26	NM	STRIP4-F	J3	Connector 4 Pin		NM	Top	PTH	1
27	NM	STRIP6	J4, J6	Connector 6 Pin		NM	Top	PTH	2
28	HEADER 5X2	M20-78505XX	J7, J9, J10	PTH Header, 5x2, Pitch 2.54mm, Female Bottom Entry		CN-M20-7850546	Bottom	PTH	3
29	CON2		J11, J12	Phoenix Connector, Male, P=3.81mm, 90°, 2 Pin		CN-PH1803277	Top	PTH	2
30	CON3	SAURO3M-P3_81_90	JA, JB	Phoenix Connector, Male, P=3.81mm, 90°, 3 Pin		CN-PH1803280	Top	PTH	2
31	JUMPER	SIP-2	JMP1, JMP8, JMP10, JMP11, JMP13, JMP18, JMP20, JMP21, JMP22	Jumper, 2 Pin, Pitch 2.54mm		CN-STRL5169140231116	Top	PTH	9
32	NM	SIP-2	JMP2, JMP5, JMP12, JMP15, JMP16, JMP17, JMP23	Jumper		NM	Top	PTH	7
33	JUMPER3	SIP-3	JMP3, JMP4	Jumper, 3 Pin, Pitch 2.54mm		CN-STRL5169140231116	Top	PTH	2
34	47n	E1008L	L1, L2, L6, L7, L11	Inductor Murata, 47nH, LQW2UAS47NJ00L		IN-LQW2UAS47NJ00L	Top	SMD	5
35	470n	E1008L	L3, L4, L5, L9	Inductor Murata, 470nH, DFE252012PD-R47M		IN-DFE252012PD-R47M	Top	SMD	4
36	Green	LED0603	LD1, LD2, LD3, LD4, LD5, LD6, LD7, LD13, LD14	Green LED, 0603		OP-SMD0603VERDE	Top	SMD	9
37	GREEN	LED0603	LD8, LD10	Green LED, 0603		OP-SMD0603VERDE	Bottom	SMD	2
38	LightPipe	PIPE_SLP3-150-250-R	LP1, LP2, LP5, LP6	Light Pipe, SLP3-150-250-R		OP-SLP3-150-250-R	Top	PTH	4
39	IRLML2502	E_SOT23	Q1, Q2	N-Channel MOSFET, IRLML2502		TR-IRLML2502PBF	Top	SMD	2
40	STN3N40K3	SOT-223	Q7, Q10	N-Channel Power-MOSFET, STN3N40K3		TR-STN3N40K3	Top	SMD	2
41	0R	E0603R	R1, R8, R19, R20, R28, R29, R36, R50, R55, R71, R82, R93, R94, R101, R141, R147, R148, R149, R158, R161, R162, R163, R174, R175, R176, R177, R183, R184, R186, R187, R200, R207, R208, R213, R214, R223, R229, R230	Resistor 0603		RE-00603	Top	SMD	38
42	100R	E0603R	R2, R9, R10, R11	Resistor 0603, 5%		RE-1010603	Bottom	SMD	4
43	1K	E0603R	R3, R24	Resistor 0603, 5%		RE-10206035%	Bottom	SMD	2
44	100R	E0603R	R4, R16, R17, R205	Resistor 0603, 5%		RE-1010603	Top	SMD	4
45	1K	E0603R	R5, R7, R15, R25, R74, R88, R114	Resistor 0603, 5%		RE-10206035%	Top	SMD	7
46	NM	E0603R	R6, R47, R48, R66, R75, R78, R80, R83, R92, R122, R143, R144, R145, R146, R180, R189	Resistor 0603, 5%, Resistor 0603, Resistor 0603, 1%		NM	Top	SMD	16
47	0R	E0603R	R12, R14, R18, R23, R27, R46, R150, R152, R181, R210, R212, R219, R220, R235, R236	Resistor 0603		RE-00603	Bottom	SMD	15
48	33K	E0603R	R13, R185	Resistor 0603, 5%		RE-3330603	Top	SMD	2
49	10K	E0603R	R21, R22, R40	Resistor 0603, 5%		RE-10306035%	Top	SMD	3
50	100K	E0603R	R26, R30, R72, R111, R115, R116, R124, R126	Resistor 0603, 1%		RE-10406031%	Top	SMD	8
51	330K	E0603R	R31, R41	Resistor 0603, 5%		RE-3340603	Top	SMD	2

52	2K2	E0603R	R32, R33, R65, R110, R113, R154, R178, R179, R182	Resistor 0603, 1%		RE-22206031%	Top	SMD	9
53	680R	E0603R	R34, R42	Resistor 0603, 5%		RE-6810603	Top	SMD	2
54	51K1	E0603R	R35, R43	Resistor 0603, 1%		RE-51K10603	Top	SMD	2
55	82K	E0603R	R37, R51	Resistor 0603, 1%		RE-82306031	Top	SMD	2
56	24K9	E0603R	R38, R52	Resistor 0603, 1%		RE-24K96030	Top	SMD	2
57	510R	E0603R	R39, R59	Resistor 0603, 1%		RE-51106031	Top	SMD	2
58	22K	E0603R	R44, R57, R123	Resistor 0603, 1%		RE-22306031	Top	SMD	3
59	10R	E0603R	R53, R54, R56, R58	Resistor 0603, 5%		RE-1000603	Top	SMD	4
60	10K	E0603R	R60, R68, R70, R97, R108, R109, R125, R190	Resistor 0603, 1%		RE-1030603	Top	SMD	8
61	3K9	E0603R	R61	Resistor 0603, 1%		RE-39206031	Top	SMD	1
62	1K2	E0603R	R62	Resistor 0603, 1%		RE-12206031	Top	SMD	1
63	33R	E0603R	R63, R96, R117	Resistor 0603, 1%		RE-3300603	Bottom	SMD	3
64	220K	E0603R	R64, R77, R95	Resistor 0603, 1%		RE-22406031%	Top	SMD	3
65	1K09	E0603R	R67	Resistor 0603, 1%		RE-1K090603	Top	SMD	1
66	18K	E0603R	R69, R81, R188	Resistor 0603, 1%		RE-1830603	Top	SMD	3
67	680R	E0603R	R73	Resistor 0603, 1%		RE-68106031	Top	SMD	1
68	120R	E1206R	R76, R79, R102, R103	Resistor 1206, 1%		RE-12112061	Top	SMD	4
69	1R5	E1206R	R84, R104	Resistor 1206, 1%		RE-1R512061	Top	SMD	2
70	1R	E1206R	R85, R99, R100, R112, R151, R156, R157, R160, R172, R173	Resistor 1206, 1%		RE-1R12061%	Top	SMD	10
71	10R	E1206R	R86, R128	Resistor 1206, 1%		RE-10R1206	Top	SMD	2
72	12R	E1206R	R87, R129	Resistor 1206, 1%		RE-12R12061	Top	SMD	2
73	220K	E0805R	R89, R90, R130, R131	Resistor 0805, 1%		RE-2240805	Top	SMD	4
74	1K	E0603R	R91, R118, R119, R120, R142, R153	Resistor 0603, 1%		RE-1020603	Top	SMD	6
75	120K	E0603R	R98, R106, R107	Resistor 0603, 1%		RE-12406031	Top	SMD	3
76	8K2	E0603R	R105, R121	Resistor 0603, 5%		RE-8220603	Top	SMD	2
77	4K87	E0603R	R127	Resistor 0603, 1%		RE-4K871%0603	Top	SMD	1
78	10K	E0603R	R132, R134, R137, R194	Resistor 0603, 5%		RE-10306035%	Bottom	SMD	4
79	2K2	E0603R	R133, R135	Resistor 0603, 1%		RE-22206031%	Bottom	SMD	2
80	NM	E0603R	R139, R195	Resistor 0603, 5%		NM	Bottom	SMD	2
81	100K	E0603R	R155	Resistor 0603, 5%		RE-1040603	Top	SMD	1
82	100K	E0805R	R165, R203	Resistor 0805, 5%		RE-1040805	Top	SMD	2
83	47K	E1206R	R167, R169, R198, R202	Resistor 1206, 5%		RE-4731206	Top	SMD	4
84	220K	E1206R	R191, R197	Resistor 1206, 5%		RE-2241206	Top	SMD	2
85	4K7	E0603R	R192	Resistor 0603, 1%		RE-47206031%	Top	SMD	1
86	RT424005	FINDER4052	RL1, RL2, RL3, RL4, RL9, RL10	TE Relay, 2 Contacts, 5VDC, RT424005		RL-RT424005	Top	PTH	6
87	IMC03GR	IMC0XGR	RL5, RL6, RL7, RL8	TE Relay, Single Contact, 5VDC, IMC03GR		RL-IMC03GR	Top	SMD	4
88	0R015	E0805R	SH1	Shunt Resistor 0805, 1%		RE-CSS0805FT15L0	Top	SMD	1
89	760390011	760390011	T1, T2	Würth Transformer, 760390011		IN-760390011	Bottom	SMD	2
90	750313638	WE-PPT1_1209_L12.7W9.14H 7.62	TF1, TF2, TF3	Würth Transformer, 750313638		TF-750313638	Top	SMD	3
91	XC7A35T-1FTG256C	FTG256	U1	Xilinx Artix-7 FPGA, 256BGA, XC7A35T-1		IC-XC7A35T-1FTG256C	Top	SMD	1
92	LT3042	MSOP10G	U2, U14	Analog D. Low Noise Linear Reg., LT3042IMSE#PBF		IC-LT3042IMSE#PBF	Top	SMD	2
93	IS25WP128	SO-8W	U3	ISSI 128Mb Flash, IS25WP128F-JBLE		IC-IS25WP128	Top	SMD	1
94	ISO7310C	SO-8	U4, U25	TI Single Channel Digital Isolator, ISO7310CDR		IC-ISO7310CDR	Bottom	SMD	2
95	LP2981-5V	SOT23-5	U5, U28	TI 100mA Ultralow-Dropout Regulator,LP2981IM5-5.0		IC-SMDLP2981AIM5-5.0	Bottom	SMD	2
96	TPS72301	SOT23-5	U6, U12	TI Low-Noise Negative LDO Regulator		IC-TPS72301DBVR	Top	SMD	2
97	74LVC1G14	SOT353	U7	Single Schmitt-trigger inverter, 74LVC1G14		IC-SMD74LVC1G14GW	Bottom	SMD	1
98	SN6507-Q1	HVSSOP10	U8, U10, U37	TI Transformer Driver for Isolated Power Supply		IC-SN6507DGQRQ1	Top	SMD	3
99	TPS628501-Q1	SOT583	U9, U11, U13, U21	TI Step-Down Converter, TPS628501QDRLRQ1		IC-TPS628501QDRLRQ1	Top	SMD	4
100	OPA197	SOT23-5	U15, U16	TI Rail-To-Rail Low Offset Voltage Op. Amp.,OPA197		IC-OPA197IDBVT	Top	SMD	2
101	ULN2003	SO-16	U17, U30	Darlington Array, 7 NPN, ULN2003AD		IC-SMDULN2003D SMD	Top	SMD	2
102	NM(DS90LV001)	SO-8	U18	TI 800Mbps LVDS Buffer, DS90LV001TM		NM	Top	SMD	1
103	24AA025	SO-8	U19	Microchip 2K EEPROM w/ Unique 32-bit Serial Number		IC-24AA025UID-I/SN	Top	SMD	1
104	74LVC1G08	SOT23-5	U20	TI Single 2-Input AND Gate, SN74LVC1G08DBVR		IC-SN74LVC1G08DBVR	Top	SMD	1
105	74LVC00	SO-14	U22	Quad 2-Input NAND Gate, 74LVC00AD		IC-SMD74LVC00AD	Top	SMD	1
106	STM32G030F6P6	TSSOP20	U23, U26	ARM Cortex-M0 32b MCU, 32KB Flash		IC-STM32G030F6P6	Top	SMD	2
107	74AHC1G125	SOT23-5	U24	TI Single Bus Buffer Gate With 3-State Output		IC-SN74AHC1G125DBVR	Top	SMD	1
108	PCA9534	TSSOP16	U27	TI 8-bit I2C IO Expander, PCA9534PWR		IC-PCA9534PWR	Top	SMD	1
109	18024115401H	SO-16W	U29	Würth 4-Channel Digital Isolator w/ DC-DC Module		IC-18024115401H	Bottom	SMD	1
110	SN6505B	SOT23-6	U31, U38	TI 1A Transformer Driver, SN6505BDBVR		IC-SN6505BDBVR	Bottom	SMD	2
111	AD7405	SO-16WL	U33, U35	Analog D. Isolated Sigma-Delta Modulator w/ LVDS		IC-AD7405BRIZ	Top	SMD	2
112	AD7405	SO-16WL	U34, U36	Analog D. Isolated Sigma-Delta Modulator w/ LVDS		IC-AD7405BRIZ	Bottom	SMD	2
113	100MHz	VCXO_2.5x2	Y1	Abracon Low-Jitter LVDS Oscillator, 100MHz		XT-AK2BDDF1-100.0000T	Top	SMD	1
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